

A Meta-Analytic Review of the Relationship Between Epistemic Cognition and Academic Achievement

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Epistemic cognition, defined as the ways that people acquire, justify, and use knowledge, has been a prominent area of scholarship in educational psychology for nearly 50 years. Researchers have argued that epistemic cognition is a key predictor of many 21st century learning outcomes including critical thinking, scientific literacy, and historical thinking, among others. Despite a large volume of quantitative empirical research on epistemic cognition and academic achievement, there has been no published systematic analysis of this literature. We conducted a meta-analysis of 132 nonexperimental studies in this literature, and found epistemic cognition, as measured predominantly in terms of beliefs, was positively correlated with academic achievement, $r = .162$, $p < .001$, indicating a small but meaningful relationship. Moderator analyses revealed models and instruments focusing on development and justification of knowledge had higher correlations with academic achievement than those focused on constructs related to authority. We found evidence supporting domain-general, domain-specific, and topic-specific conceptualizations of epistemic cognition, and stronger correlations when the specificity of the epistemic cognition measure matched the specificity of the achievement measure. Conceptual knowledge and argumentation performance were more strongly associated with epistemic cognition than declarative or procedural knowledge performance. Finally, we found epistemic cognition predicted academic achievement from elementary school through graduate school, suggesting the importance of studying epistemic cognition across all educational levels. These findings have direct implications for honing epistemic cognition theory, creating better measures of epistemic cognition, and developing effective educational interventions on this critical topic.

Educational Impact and Implications Statement

The findings from this meta-analysis show a small but important relationship between epistemic cognition and academic achievement. Epistemic cognition seems most strongly related to conceptual understanding and argumentation in academic domains. Therefore, educators who wish to help their students develop such knowledge and skills should investigate ways of fostering students' epistemic cognition. Our study points to several key foci for such work, including teaching students the ways knowledge develops and is justified within academic domains.

Keywords: epistemic cognition, epistemic beliefs, academic achievement, personal epistemology, meta-analysis

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The 21st century has ushered in an era of unprecedented expansion of information production and access, resulting in increasing public and scholarly focus on that information's quality and veracity (Leu, Kinzer, Coiro, Castek, & Henry, 2013; OECD, 2013). Productive participation in everyday life, and indeed democracy itself, requires people who can discern accurate and useful contributions to human knowledge from speculation, dogma, and intentional decep-

tions (Rapp, 2016; Sandoval, Greene, & Bråten, 2016). To prepare students for such work, educators have revised curricular standards and goals to encompass a myriad of higher order processes including critical thinking (Bonney & Sternberg, 2011; National Education Association, 2014) and argumentation (National Governors Association Center for Best Practices, 2010), as well as discipline-specific knowledge and practices including scientific literacy (Bromme & Goldman, 2014; Elby, Macrander, & Hammer, 2016; NGSS Lead States, 2013), historical understanding (VanSledright & Maggioni, 2016; Wineburg, 1991), and mathematical understanding (Depaepe, De Corte, & Verschaffel, 2016; Weber, Inglis, & Mejia-Ramos, 2014). Unfortunately, such higher order processing is far from natural or typical (Sinatra, Kienhues, & Hofer, 2014; Stanovich, 2010), and there is a clear need for continued research into how to help people more effectively critique and produce knowledge so that they can engage productively within the modern world (Greene & Yu, 2016;

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Hofer & Pintrich, 1997; King & Kitchener, 1994; Kuhn, Cheney, & Weinstock, 2000). Such higher order processing requires epistemic cognition, which Greene, Sandoval, and Bråten (2016a) defined as “how people acquire, understand, justify, change, and use knowledge in formal and informal contexts” (p. 1).

The volume of empirical research on epistemic cognition has increased exponentially in last 50 years, revealing relations between epistemic cognition and performance in numerous contexts including, among others, reading comprehension (Bråten, Britt, Strømsø, & Rouet, 2011), psychology (Hofer, 2000), mathematics (Muis, 2004; Schommer, Crouse, & Rhodes, 1992), history, (Maggioni, VanSledright, & Alexander, 2009), science (Barzilai & Weinstock, 2015; Hammer, 1994; Sandoval, 2005), public understanding of science (Greene, Yu, & Copeland, 2014), Internet searching (Kammerer, Amann, & Gerjets, 2015), and out-of-school learning (Bricker & Bell, 2016). However, there remain ongoing concerns regarding the many differing conceptualizations of epistemic cognition (Hofer & Bendixen, 2012; Hofer & Pintrich, 1997; Sandoval, 2009; Schommer-Aikins, 2004), the degree to which epistemic cognition is or is not domain-general, domain-specific, or even topic-specific (Bråten, Strømsø, & Samuelstuen, 2008; Muis, Bendixen, & Haerle, 2006), the psychometric properties of quantitative epistemic cognition measures (DeBacker, Crowson, Beesley, Thoma, & Hestevold, 2008; Greene & Yu, 2014; Hofer & Sinatra, 2010), and the variability in empirical findings across studies and contexts (Sandoval et al., 2016). Such concerns are not surprising given the rapid proliferation of research and ideas in the field; iterative expansion, testing, and pruning of conceptualizations and models are indicative of a healthy, vigorous area of scholarship (Greene, Sandoval, & Bråten, 2016b; Murphy, Alexander, & Greene, 2011). Nonetheless, such concerns must be addressed to determine how best to (a) understand and conceptualize epistemic cognition, (b) measure people’s epistemic cognition, and (c) design interventions to foster critical consumption and production of knowledge in the 21st century (Bråten, 2016).

Given ongoing concerns about the nature and measurement of epistemic cognition, we felt the field was overdue for a meta-analysis of the relationship between epistemic cognition and academic achievement. Therefore, the purpose of our study was to conduct a quantitative meta-analysis of the empirical, nonexperimental literature on the relationship between epistemic cognition and academic achievement. We chose to focus on nonexperimental studies because the majority of epistemic cognition research has not involved interventions, and because we felt the conceptual and methodological debates in the field warranted a synthesis of studies that did not involve some kind of manipulation of the target variables or participants, which would have added variance to the relations of interest.

To investigate these conceptual and methodological debates, we investigated how the relationship between epistemic cognition and academic achievement varied by both theoretical (e.g., underlying model of epistemic cognition, instrument used, degree of domain-specificity posited) and methodological (e.g., peer-reviewed vs. non-peer-reviewed, reliability of the measure) factors. These moderator analyses were used to investigate ongoing concerns in the field regarding the most useful conceptualizations and operationalizations of epistemic cognition (e.g., the degree of domain-generality of epistemic cognition, or which constructs are and are not epistemic in nature; Greene et al., 2016b; Hofer & Bendixen,

2012; Sandoval et al., 2016). Such work is needed to guide the next 50 years of research in this critical area of scholarship.

Research on Epistemic Cognition

Models of Epistemic Cognition

There are a number of epistemic cognition models that share common roots but have also diverged substantially over time (Sandoval et al., 2016). Foundational models of epistemic cognition, previously called personal epistemology models (Hofer & Pintrich, 1997), generally fall into one of four categories: developmental, dimensional, philosophically-informed, or academic-discipline-informed models. Over time, researchers have tested, revised, and integrated these models. Importantly, despite the relatively recent move to broadly conceiving of epistemic cognition as encompassing not only beliefs but also the practices and other cognitive and metacognitive processes dynamically interacting with those beliefs during knowledge construction and critique (Barzilai & Zohar, 2014; Chinn, Buckland, & Samarapungavan, 2011; Greene et al., 2016a; Sinatra, 2016), most of the studies we found for this meta-analysis of epistemic cognition focused on beliefs rather than practices or processes. There is emerging scholarship on broader views of epistemic cognition encompassing more than just beliefs (cf. Sinatra, 2016), but the majority of those studies have involved qualitative analyses (e.g., Herrenkohl & Cornelius, 2013).

Developmental models. Perry (1970) pioneered the field of personal epistemology through a study of male college students’ ethical and intellectual development. According to Perry’s model, individuals develop through sequential stages characterized by evolving views about knowledge and knowing that cut across all domains and experiences. Developmental shifts across stages are initiated through cognitive disequilibrium (Sandoval et al., 2016) and involve the coordination of subjective and objective dimensions of knowing (Kuhn et al., 2000). People start in the dualist stage, in which they have an objective view of the world; knowledge is certain, static, and comes from an external, authoritative source. Over time, individuals can adopt a multiplistic, wholly subjective view in which all assertions are opinions, knowledge is uncertain, and thus critical thinking is useless. The relativist stage is achieved when individuals understand that knowledge is conditional and varies across different contexts, and that knowing requires logic and reasoning. In Perry’s final stage, commitment within relativism, an individual’s views of knowledge and knowing are shaped by the values and identities they espouse.

Others have built their own models from Perry’s work (Baxter Magolda, 1992; King & Kitchener, 1994, 2004; Kuhn et al., 2000), proposing similar stages, stances, or views of epistemic cognition. Kuhn et al. (2000) proposed a model of epistemological thinking in which people progressively adopt increasingly sophisticated views regarding the objective and subjective aspects of knowledge, but unlike Perry, they posited that an individual’s stage of epistemological thinking can vary across different judgment domains (e.g., aesthetics, value). The first two stages in Kuhn et al.’s (2000) model, absolutist and multiplist, mirror the dualist and multiplist stages of Perry’s (1970) model. In the final stage of Kuhn et al.’s model, the objective and subjective dimensions of knowing are reconciled through an evaluativist perspective. People in this stage

view knowledge as uncertain but believe critical thinking can be used to evaluate evidence and decide between knowledge claims. As people progress through the stages of developmental models, their epistemic cognition becomes more advanced and useful for learning. Researchers have investigated these developmental trajectories, and found relations between stages and learning outcomes (e.g., Kuhn et al., 2000; Mason & Scirica, 2006).

Dimensional models. The introduction of multidimensional beliefs models marked a significant shift in the field of epistemic cognition (Hofer & Pintrich, 1997; Schommer, 1990). According to these models, individuals vary on multiple, mostly independent dimensions of beliefs about the nature of knowledge and the nature of knowing, which lie on a continuum from naïve to sophisticated. Whereas developmental models were based on the idea that one's epistemic cognition develops systematically across a set of characteristics defining each stage, authors of dimensional models have claimed that development may occur asynchronously across multiple dimensions.

Schommer (1990) proposed the first multidimensional model of epistemic cognition, which included simple knowledge, certain knowledge, omniscient authority, innate ability, and quick learning dimensions. Hofer and Pintrich (1997) referred to the first two dimensions as beliefs about knowledge (i.e., that knowledge is simple or complex; that knowledge is certain or tentative). The omniscient authority dimension captured how individuals view the source of knowledge; either as handed down by authority or derived from reason and the self. Hofer and Pintrich (1997) classified innate ability (i.e., one's ability to learn is innate rather than developed) and quick learning (i.e., learning occurs quickly, or not at all) as beliefs about the nature of learning.

Hofer and Pintrich (1997) reviewed prior developmental and dimensional models of epistemic cognition, comparing and contrasting their features, and subsequently developed their own model of what they called epistemological theories. Similar to Schommer's (1990) model, simple knowledge and certain knowledge were identified as separate dimensions about the nature of knowledge. Hofer and Pintrich's source of knowledge dimension encompassed beliefs that knowledge is derived from authority to beliefs that knowledge can be constructed by the self, which was akin to Schommer's omniscient authority dimension. The final dimension in Hofer and Pintrich's model, justification of knowing, ranged from the naïve view that justification is based on appeals to authorities, to the advanced view that justification is based on inquiry norms and evaluation of the source (Hofer, 2004). Dimensional models have led to numerous quantitative investigations of relations between epistemic cognition and achievement due to the ease with which the dimensions can be operationalized into surveys and other self-report instruments (Mason, 2016).

Models informed by philosophy. In several contemporary models of epistemic cognition, researchers have borrowed from the field of philosophical epistemology to inform their models, particularly epistemology's conceptualization of justification. For example, using Royce's (1959) model of psychological epistemology, which is grounded in philosophy and psychology, Muis (2008) has studied relations among learning and three epistemic profiles that differ on how knowledge is derived and justified. These profiles are thought to guide the kinds of cognitive processes people invoke when evaluating knowledge. Rationalists primarily use reason and logic to derive and justify knowledge,

empiricists use direct observation, and metaphorists use intuition and evaluations of universality. Muis's model, based on Royce's profiles, has been used to evaluate the consistency hypothesis, which suggests people will attend most to knowledge claims supported via the cognitive processes implied by their Royce profile (e.g., Muis, Kendeou, & Franco, 2011).

Greene, Azevedo, and Torney-Purta (2008) proposed a model that integrated developmental and dimensional models of epistemic cognition with concepts from philosophy and theory of mind literatures. Given the emphasis on justification in classical philosophical epistemology, the authors proposed the expansion of Hofer and Pintrich's (1997) justification of knowing dimension to include justification by authority and justification by personal experience. In Greene et al.'s (2008) model, individuals develop through general stages, with each stage characterized by a pattern of responses across multiple dimensions of beliefs.

Bråten et al. (2011) proposed a theoretical model of epistemic beliefs built on many of their empirical studies of how students read, comprehend, and integrate multiple texts to generate knowledge or accept a claim. Like dimensional models, there were four dimensions categorized as beliefs about the nature of knowledge (i.e., simplicity of knowledge, certainty of knowledge) or the nature of knowing (i.e., source of knowledge, justification of knowledge). Following Greene et al.'s (2008, 2010) argument about the primacy of justification in epistemology, the authors expanded the justification dimension to be comprised of both justification by personal experience and justification by authority. Through analysis of think-aloud protocols, Ferguson, Bråten, and Strømsø (2012) found support for a third dimension of justification: multiple justification. This dimension has been integrated into their model and subsequent research.

Chinn et al. (2011) challenged developmental and dimensional models by positing epistemic cognition as a situated, social phenomenon. In the AIR model of epistemic cognition (Chinn, Rinehart, & Buckland, 2014), there are three broad categories of epistemic cognition: epistemic aims, epistemic ideals, and reliable processes. Epistemic aims are the intended goals of cognition to achieve an epistemic end such as knowledge, understanding, or true beliefs. The greater value people assign to an epistemic end, the more likely it is they will set epistemic aims to achieve it. Epistemic ideals are the standards or criteria that must be met to accept an epistemic product. For example, an individual might decide that a scientific knowledge claim is acceptable only if it is supported by empirical evidence. Processes used to produce knowledge can be more or less reliable, and their reliability is dependent on certain conditions. Chinn et al.'s broader conceptualization of epistemic cognition was designed to capture students' contextualized ideas about their production of knowledge, that is, their "practical epistemologies" (Sandoval, 2005).

Models informed by academic disciplines. Some researchers have proposed models of epistemic cognition with strong disciplinary grounding (e.g., Elby & Hammer, 2010; Hammer, 1994; Lee, Goldman, Levine, & Magliano, 2016; Weber et al., 2014). The authors of these models have based their work on research involving learning and instruction in particular academic domains or disciplines and often emphasize the knowledge and mechanisms of the discipline, rather than broad beliefs about the nature of knowledge or knowing. For example, Hammer (1994) developed a framework of epistemic resources regarding the co-

herence and content of physics knowledge. Elder (1999) synthesized the research on science-specific epistemic beliefs and identified seven key ideas that represented the nature of knowledge and knowing in science, such as the role of evidence and experiments. Hammer and Elby (2003) challenged the notion of stable, decontextualized, overarching “epistemic beliefs” and argued for the term “epistemological resources” to emphasize the fine-grained components of an individual’s epistemic cognition that are made active in relation to specific contexts.

Summary of models of epistemic cognition. In sum, there has been some convergence of epistemic cognition models (Greene et al., 2016b), but scholars continue to construct new conceptualizations of epistemic cognition (e.g., Barzilai & Zohar, 2014). Therefore, it seemed warranted to investigate whether the relationship between epistemic cognition and academic achievement varies by the underlying conceptual framework used. Such findings might help to address concerns about the large number of models in the field, and whether the field would benefit from greater precision in, and perhaps pruning of, definitions and operationalizations of key constructs (Alexander, 2016).

Finally, over the course of coding studies for this meta-analysis, we found a group of them that, despite references to epistemic

cognition, appeared to have no clear grounding in any of the models described in this section (e.g., Tu, Shih, & Tsai, 2008). Instead, these studies seemed much more grounded in theories of constructivism (i.e., humans create their own mental representations and understandings of the world; Phillips, 1995). Therefore, we coded these studies as adhering to a constructivist model, to determine via moderator analysis whether such studies’ effect sizes differed from studies utilizing more traditional epistemic cognition models. If indeed epistemic cognition differs from ideas of constructivism (cf. Hofer & Pintrich, 1997), then studies purporting to be about epistemic cognition but including constructivist framing or factors may be skewing the literature.

Measurement of Epistemic Cognition

To date, the most common way researchers have investigated epistemic cognition is through self-report questionnaires. The majority of these measures assess each construct using multiple statements or items with Likert-type response scales. See Table 1 for an overview of major epistemic cognition instruments, response scales, constructs, and example items. Schommer’s (1990) development of the Epistemological Questionnaire (EQ), which

Table 1
List of Authors and Epistemic Cognition Instruments

Authors	Instrument	Directions/scale	Epistemic cognition constructs or levels	Example item
Schommer (1990)	Epistemological Questionnaire (EQ)	Given one statement, choose agreement on 5-point Likert scale (1 = <i>strongly disagree</i> ; 5 = <i>strongly agree</i>)	Simple knowledge; certain knowledge; innate ability; quick learning; omniscient authority	I don’t like movies that don’t have an ending.
Schraw, Dunkle, & Bendixen (1995)	Epistemic Belief Inventory (EBI)	Given one statement, choose agreement on 5-point Likert scale (1 = <i>strongly disagree</i> ; 5 = <i>strongly agree</i>)	Simple knowledge; certain knowledge; omniscient authority; innate ability; quick learning	When someone in authority tells me what to do, I usually do it.
Hofer (2000)	Discipline-Focused Epistemic Belief Questionnaire (DEBQ)	Given one statement, choose agreement on 5-point Likert scale (1 = <i>strongly disagree</i> ; 5 = <i>strongly agree</i>)	Certainty/simplicity of knowledge; personal justification for knowing; authority as source of knowledge; and attainment of truth	Truth in this subject is unchanging.
Kuhn, Cheney, & Weinstock (2000)	Epistemological Understanding Instrument (EUI)	Given two statements, can only one of their views be right, or could both have some rightness? Choose one option: Only one can be right or both could have some rightness. If both could be right, choose one option: one could be more right or one could not be more right than the other	Realist; absolutist; multiplist; evaluator	Robin believes one book’s explanation of how the brain works. Chris believes another book’s explanation of how the brain works.
Conley, Pintrich, Vekiri, & Harrison (2004)	Conley et al.’s epistemological beliefs in science questionnaire	Given one statement, choose agreement on 5-point Likert scale (1 = <i>strongly disagree</i> ; 5 = <i>strongly agree</i>)	Source of knowledge; certainty of knowledge; development of knowledge; justification of knowledge	All questions in science have one right answer.
Ferguson, Bråten, Strømsø, & Anmarkrud (2013)	Justification for Knowing Questionnaire (JFK-Q)	Given one statement, choose agreement on 10-point Likert scale (1 = <i>disagree completely</i> ; 10 = <i>agree completely</i>)	Justification by authority; personal justification; justification by multiple sources	Things that are written in natural science textbooks are correct.

measured the five dimensions in her model, initiated quantitative investigations of epistemic cognition and a sharp increase in research exploring the construct. Schraw, Bendixen, and Dunkle (2002) created a similar instrument informed by Schommer's (1990) model of epistemological beliefs, the Epistemic Belief Inventory (EBI), which measured the same five dimensions.

Hofer (2000) investigated the question of domain-generalizability of epistemic cognition by measuring the epistemic beliefs of first-year college students across two disciplines, psychology and science. She developed the Discipline-Focused Epistemic Belief Questionnaire (DEBQ), based on Hofer and Pintrich's (1997) model, and administered the DEBQ alongside the domain-general EQ. Factor analyses of the DEBQ revealed four latent factors: certainty/simplicity of knowledge, personal justification for knowing, authority as source of knowledge, and attainment of truth. Hofer (2000) found statistically significant differences in students' epistemic beliefs across disciplines, supporting the notion that epistemic cognition can be domain or discipline-specific.

To measure an individual's stage of epistemological understanding, Kuhn et al. (2000) developed the Epistemological Understanding Instrument (EUI). This 15-item instrument included questions targeting the transitions in epistemological thinking between stages, taking into account different judgment domains such as value judgments, aesthetic judgments, and judgments of truth about the physical world. Participants were asked to judge a pair of statements on whether only one claim could be right, representing absolutist thinking, or that both could have some rightness, representing multiplist thinking. If an individual responded that both could be right, they were then asked if one statement could be more correct than the other, with affirmative responses interpreted as evidence of evaluativism.

Conley, Pintrich, Vekiri, and Harrison (2004) created a questionnaire to measure elementary school students' epistemological beliefs in science. The researchers based their instrument on Hofer and Pintrich's (1997) model, and found four dimensions regarding the nature of knowledge (i.e., source, certainty) and the nature of knowing in science (i.e., development, justification). The development of knowledge factor was defined as "a belief that recognizes science as an evolving subject and that ideas and theories can change on the basis of new data and evidence" (p. 190). Conley et al.'s (2004) research contradicted previously published claims that young students could not have advanced epistemological thinking. Focusing on measurement of individuals' domain-specific epistemic beliefs about justification, Ferguson et al. (2013) created the Justification for Knowing Questionnaire (JFK-Q), which assessed how individuals justify knowing in science across three dimensions: justification by authority, personal justification, and justification by multiple sources. Barzilai and Weinstock (2015) created the Epistemic Thinking Assessment (ETA) to measure individual's topic-specific, epistemic beliefs on multiple dimensions using items similar to the EUI (Kuhn et al., 2000). Using the individual's item responses, researchers could classify individual's beliefs within the developmental stages proposed by Kuhn et al. (2000).

Survey and self-report measures have been scrutinized due to poor reliability and construct validity (Clarebout, Elen, Luyten, & Bamps, 2001; DeBacker et al., 2008; Hofer & Sinatra, 2010; Mason, 2016). Researchers continue to develop new self-report measures of epistemic cognition. Also, researchers have explored other ways to measure epistemic cognition, including think-aloud protocols, discourse analysis, observations, and reflective inter-

views (Mason, 2016), but in many cases, those measures do not result in scores amenable to quantitative analysis. Given the variety of instruments used to capture epistemic cognition, there is a need to determine whether the relationship between epistemic cognition and academic achievement varies based upon the instrument used.

Constructs Included in Epistemic Cognition Measures

Hofer and Pintrich (1997) were able to synthesize much of the epistemic cognition literature published to that point by grouping the various constructs into three categories: the nature of knowledge (e.g., simple knowledge, certain knowledge), the nature of knowing (e.g., source of knowledge, justification of knowledge), and the nature of learning (e.g., quick learning, innate ability). Many studies of epistemic cognition have included some version of these constructs, ranging from strict adherence to these construct names and definitions (e.g., Hofer, 2000) to subtle changes. A good example of the latter is the Connotative Aspects of Epistemological Beliefs (CAEB; Stahl & Bromme, 2007) measure, which included a construct called "texture" that measured beliefs about the structure and accuracy of knowledge (i.e., a combination of simple and certain knowledge constructs), and a construct called "variability" that tapped beliefs about stability of knowledge (i.e., a variation of the certain knowledge construct).

Debates continue about what constructs should be considered "epistemic" versus "non-epistemic" in nature, with the nature of learning constructs receiving a great deal of scrutiny (Elby, 2009; Greene et al., 2008; Hofer & Pintrich, 1997; Sandoval, 2009). Many researchers have proposed additional constructs beyond those categorized by Hofer and Pintrich (1997), including development of knowledge (Conley et al., 2004), various factors involving how people justify knowledge qua knowledge (Bråten et al., 2011; Greene et al., 2008; Murphy, Alexander, Greene, & Edwards, 2007), epistemic aims (Chinn et al., 2011), and a wide variety of factors aligned with constructivism (e.g., Gonida & Metallidou, 2015). We coded for these constructs so that we could investigate whether some factors were better predictors of academic achievement than others, which might inform debates about what factors are "epistemic" or not, as well as which are promising directions for continued research, versus those that are not.

Nature of learning constructs. Schommer-Aikins' (2004) quick learning and innate ability factors, and the items typically used to measure them (e.g., "The really smart students don't have to work hard to do well in school" [Schommer, 1990, p. 500]), seem more closely tied to Dweck's (1999) implicit theories of intelligence or Schoenfeld's (1992) work on mathematical beliefs than they do to the production and use of knowledge claims. Schommer-Aikins (2004) appeared to endorse Hofer and Pintrich's (1997) categorization of these factors as not epistemic but rather focused on the nature of learning, but nonetheless researchers continue to include these factors in their conceptualizations and measurements of epistemic cognition (e.g., Hotulainen & Telivuo, 2014; Oh, Chung, Han, Woo, & Deiter, 2016; Ricco, Pierce, & Medinilla, 2010). There are numerous studies revealing strong empirical findings between implicit theories of intelligence and academic achievement (Dweck, 2015). Therefore, it seems important to investigate whether the strength of the relationship between epistemic cognition and academic achievement varies between the

nature of learning and other factors. It may be that the findings of strong relations between epistemic cognition and academic achievement are in some part due to the inclusion of these nature of learning factors, which might in turn call into question the utility of other epistemic cognition factors (e.g., nature of knowledge, nature of knowing).

Justification. Numerous authors have argued that given philosophical epistemology's historical focus on justification, epistemic cognition research should have a similar emphasis (Greene et al., 2008; Murphy et al., 2007). Several authors have proposed epistemic cognition factors based on various means of justification including appeals to authority and personal justification (Greene et al., 2008), and multiple justification (i.e., endorsing knowledge claims when they are supported by many different kinds of justification such as authority and personal experience; Bråten, Anmarkrud, Brandmo, & Strømsø, 2014). These authors' claims about the primacy of justification in philosophical epistemology, and therefore the importance of epistemic cognition in education, would suggest that measures of these factors should correlate highly with academic achievement.

Epistemic aims. Chinn et al. (2011, 2014, 2016) have argued that modern philosophical epistemology's scope has broadened sufficiently to warrant looking beyond justification for other relevant aspects of epistemic cognition. Empirical work on the AIR model is ongoing; we were able to find only one study involving epistemic aims and academic achievement (Dai, 2014). Nonetheless, the AIR model's influence upon the field continues to grow, thus we felt it warranted inclusion in this meta-analysis.

Constructivism. Finally, paralleling our findings regarding epistemic cognition models, our review of the relevant empirical literature revealed a set of factors that seemed to best fit the ideas of constructivism (Phillips, 1995), despite authors' claims that they were studying epistemic cognition. For example, the title of Oh et al.'s (2016) article included the term "epistemological beliefs" (p. 29) but their descriptions of items meant to capture the associated factors included "learning is a constructive process to acquire knowledge by the learner" (p. 31). Likewise, Schommer-Aikins and Easter (2006) included items such as the following in their measure of epistemic cognition: "Being a good student generally involved memorizing a lot of fact [sic]" (p. 415). These characterizations of epistemic cognition factors seemed qualitatively different than those discussed previously, and therefore we felt it necessary to categorize them using the term *constructivism* so that we could investigate whether their relations with academic achievement differed from the relations involving other constructs classified under the umbrella term of *epistemic cognition*.

Academic Domain

Chandler et al. (2002) argued that people's measured level of epistemic cognition can vary by academic domain because these domains vary in their degree of "epistemic content" (Hallett, Chandler, & Krettenauer, 2002, p. 293). For example, many laypeople view hard sciences as being more "epistemic" (i.e., objective) in nature than social sciences or the arts (Schwab, 1978). Kuhn et al. (2000) proposed that development in epistemic cognition varies by domain, with advancement in hard sciences likely occurring at a later age, or at a higher level of education, than advancement in social sciences or aesthetics. Many scholars have

found differences in participants' endorsement of epistemic cognition factors across academic domains (e.g., Buehl & Alexander, 2001; Greene, Torney-Purta, & Azevedo, 2010; Hofer, 2000). Therefore, in this meta-analysis, it seemed prudent to investigate whether the strength of the relation between epistemic cognition and academic achievement was moderated by academic domain. We also examined whether the alignment of the study's context with the achievement measure also mattered. For example, in some cases we had correlations between epistemic cognition data from a particular context (e.g., a psychology class) and academic achievement from another context (e.g., GPA; Mateos et al., 2011); such correlations were labeled as not-aligned, and compared with correlations from aligned studies (e.g., both measures designed for a science classroom context; Muis & Gierus, 2014).

Domain Generality, Specificity, and Topic-Specificity

The early literature largely characterized epistemic cognition as a domain-general phenomenon, with people's beliefs consistently affecting their functioning within and across contexts (King & Kitchener, 1994; Perry, 1970). Hofer and Pintrich (1997) argued that epistemic cognition might have both domain-general and domain-specific aspects, a view adopted by numerous other researchers in subsequent years (e.g., Barzilai & Zohar, 2014; Buehl & Alexander, 2001; Kuhn, 2000; Muis et al., 2006). As Muis et al. (2006) described in their theory of integrated domains in epistemology (TIDE) model, it may be the case that in some contexts, people's domain-general epistemic cognition exerts the strongest effect, for example, when high-school students who believe knowledge is complex and dynamic show a general willingness to engage in critical thinking across all their classes. On the other hand, this view also suggests that people have domain-specific epistemic cognition, such as when these students who believe knowledge to be complex and dynamic endorse different means of justification across contexts (e.g., relying upon authority in history, but relying on logic in science). Empirical research has revealed evidence of both domain-general consistencies and domain-specific differences in people's responses to epistemic cognition measures (Buehl & Alexander, 2001; Hofer, 2000; Muis et al., 2006).

More recently, researchers have argued that epistemic cognition can vary by topic as well (Bråten et al., 2008; Bråten, Gil, Strømsø, & Vidal-Abarca, 2009; Stahl & Bromme, 2007; Trautwein & Lüdtke, 2007). For example, within the domain of mathematics, people may believe that algebra is simple and certain, whereas geometry is complex and dynamic. Findings regarding topic-specificity have led some researchers to conclude that epistemic cognition is fundamentally situated or contextual, dependent upon dynamic interactions between people, content, tasks, and affordances of the context in which they engage in epistemic cognition (Chinn et al., 2011; Elby et al., 2016; Sandoval, 2012).

Thus, conceptualizations of epistemic cognition range across four broad categories: domain-general, domain-specific, topic-specific, and situated (Sandoval et al., 2016). Quantitative empirical research exists for the first three categories, but such research is rare for the fourth category. As such, in our meta-analysis we focused on the first three categories, to examine whether particular conceptualizations of domain-generality showed stronger relations with academic achievement than others. Further, it may be the case

that what is most important is alignment of the level of domain-specificity across both the epistemic cognition and achievement measure. From this view, as an example, domain-general epistemic cognition and a general measure of school achievement such as GPA would correlate more strongly than domain-general epistemic cognition and domain-specific academic achievement measure such as grades in a particular course. Therefore, we also examined whether studies with domain-generality-aligned measures had higher correlations than those studies with misaligned ones.

Type of Achievement Measure

Much of the argument for the importance of epistemic cognition has rested upon the idea that conceptual understanding, and the conceptual change often required for such understanding, requires effective epistemic cognition (Greene et al., 2016a; Hofer & Pintrich, 1997; Murphy et al., 2007, 2010; Sinatra & Chinn, 2011). Differences in epistemic cognition have been shown to predict conceptual change (e.g., Gill, Ashton, & Algina, 2004) as well as argumentation performance (Kuhn & Udell, 2003; Mason & Scirica, 2006). Less has been written about the role of epistemic cognition in terms of the acquisition of declarative or procedural knowledge (Greene & Yu, 2014). Therefore, it seems reasonable to investigate whether the strength of the relationship between epistemic cognition and achievement varies by type of achievement measure, with the presumption that the relationship would be stronger for conceptual and argumentation achievement measures, compared with declarative or procedural ones.

Educational Level and Epistemic Cognition

Perry (1970) posited little development in epistemic cognition prior to the college years, and many other early developmental models included similar claims (see Hofer & Pintrich, 1997). Developmental models also included assertions of a relatively fixed order of progression through each stage, roughly coinciding with both physiological maturation and exposure to education. However, empirical work has revealed far more variability, with even young children capable of displaying “sophisticated” epistemic cognition under adaptive conditions (e.g., Burr & Hofer, 2002; Conley et al., 2004; Hofer & Bendixen, 2012), middle-school and high-school students showing adept epistemic cognition (Elby & Hammer, 2010; Greene & Yu, 2014) and many adults displaying “naïve” epistemic cognition (e.g., Schommer, 1990). Such disparate findings led Chandler et al. (2002) to lament that “evidence for just about every conceivable sort of epistemic development has been shown to characterize persons of just about every conceivable age” (p. 161). Therefore, we chose to investigate level of schooling (e.g., elementary, middle, or high school, as well as college) as a moderator in our meta-analysis, expecting statistically significant correlations at each level. Failure to find such correlations in elementary school, for example, would support arguments that epistemic cognition does not develop (i.e., no variance, and thus no covariance with achievement) until later in life.

Instrument Reliability

Relations between any two constructs are necessarily constrained by the reliability of measurement of each construct; instruments with low reliability result in attenuated correlations (Crocker & Algina, 1986). Given frequently published concerns about the low reliability of measures of epistemic cognition (e.g., Barzilai & Zohar, 2014; Chinn et al., 2011; Greene et al., 2008; Muis et al., 2006), it seemed prudent to investigate whether instrument reliability moderated relations between epistemic cognition and academic achievement. A plausible hypothesis would be that an epistemic cognition instrument’s reliability would be positively related to the strength of its correlation with academic achievement.

Purpose of This Study

Given the last almost 50 years of research has led to a voluminous empirical literature on epistemic cognition, we felt the field was overdue for a meta-analysis of the relationship between epistemic cognition and academic achievement. We chose to focus on nonexperimental studies of this relationship to establish a baseline for the field. Our overall research question was:

1. What is the overall average correlation between epistemic cognition and academic achievement?

After thoroughly searching and reviewing the literature, we identified a number of theoretical and methodological factors that could plausibly moderate the relationship between epistemic cognition and academic achievement:

Theoretical moderators:

2. Does the correlation between epistemic cognition and academic achievement vary by the underlying epistemic model of the construct measured?
3. Are there particular epistemic cognition constructs that have a higher correlation with academic achievement than others?
4. Are there differences in the correlation between epistemic cognition and academic achievement across academic domains?
5. Does the correlation between epistemic cognition and academic achievement differ among domain-general, domain-specific, and topic-specific epistemic cognition instruments?
6. Is the correlation between epistemic cognition and academic achievement stronger when the domain-generality of both measures align, as opposed to not aligning?
7. Is the correlation between epistemic cognition and academic achievement stronger when the academic domain of both measures align, as opposed to not aligning?
8. Are there differences in the correlation between epistemic cognition and academic achievement across declarative knowledge, procedural knowledge, conceptual knowledge, argumentation, or general knowledge measures?

9. Are there differences in the correlation between epistemic cognition and academic achievement across types of achievement measures (i.e., standardized test, GPA, researcher developed, teacher developed, participant reported)?
10. Does the correlation between epistemic cognition and academic achievement differ across level of schooling (i.e., elementary school, middle school, high school, college)?

Methodological Moderators:

11. Do peer-reviewed sources have higher correlations between epistemic cognition and academic achievement than non-peer-reviewed sources?
12. Is the reliability of the epistemic cognition instrument positively related to the correlation between epistemic cognition and academic achievement?
13. Are there particular epistemic cognition instruments that result in higher correlations with academic achievement than others?

Method

Strategy for Searching the Literature

To discover studies of epistemic cognition and academic achievement, first we searched nine electronic databases: *PsycINFO*, *ERIC*, *Education Full Text*, and *ProQuest Dissertations & Theses Global*, as well as the *Social Sciences Citation Index*, *Arts & Humanities Citation Index*, *Conference Proceedings Citation Index—Social Science & Humanities*, *Book Citation Index—Social Sciences & Humanities*, and the *Emerging Sources Citation Index*, which are all parts of the *Web of Science Core Collection*. We used a number of search terms to explore these databases (see Table 2). As we coded studies, we conducted multiple additional searches of these databases from September 2016 through March 2017, using epistemic cognition instrument and researcher names as additional search terms.

In September and October 2016, *epistemic*, *cognition*, and *achievement* were used as the terms to search the full text of

documents in the electronic databases. We recognized that with the variety of terms that have been used in this field it would be more appropriate to use stems to search the literature. Consequently, in October through December 2016, and January through March 2017, the stems *epistem**, which matched with *epistemic* and *epistemology* and related terms; *belie** matching *belief* and *believe*; and *achiev**, for *achieve* and *achievement*, were employed in the searches. Additional terms were used to ensure that the search was exhaustive, including *performance* or *perform** instead of *achievement* and *academic*.

For the second type of search, the name or reference of an epistemic cognition instrument, or the name associated with the tool, was used to search the full text of documents in *PsycINFO*, *ERIC*, and *Education Full Text*. These terms and names included *DEBQ* and *DFEBQ* for Hofer's Domain-Focused Epistemological Beliefs Questionnaire, Schommer's *EQ*, Schraw *EBI*, and Indiana Mathematics Belief Scale, as well as Bråten, Chinn, Greene, Hofer, Muis, Schommer, and Schraw. The titles of the papers associated with the development of all 38 of the instruments in our meta-analysis were used in searches of the *Web of Science Core Collection* databases, including the *Social Sciences Citation Index*, *Arts & Humanities Index*, *Conference Proceedings Citation Index—Social Science & Humanities*, *Book Citation Index—Social Sciences & Humanities*, and the *Emerging Sources Citation Index*.

We conducted cited reference searches in the *Web of Science* for several seminal epistemic cognition studies and reviews (e.g., Hofer & Pintrich, 1997) to find additional studies. Each study that fit our inclusion criteria was also used in a *Web of Science* cited reference search. *Google* and *Google Scholar* searches were conducted in January 2017 using variable and instrument names to make sure that no appropriate studies had been missed. Additionally, the second author conducted a reference harvest of every study included in the meta-analysis. This step was also taken for several important reviews that have been written over the past 20 years. The sources cited in each chapter of the *Handbook of Epistemic Cognition* (Greene, Sandoval, & Bråten, 2016c) were also harvested. The table of contents of every publication of *Journal of Educational Psychology* and *Contemporary Educational Psychology* from 1996 through March of 2017 were also scanned to ensure no relevant studies had been missed.

Ultimately, in our electronic database searches and other investigatory methods, we discovered many studies that included epis-

Table 2
Overview of Literature Search Procedure for Electronic Databases

Literature search	Search terms	Electronic databases
September, October, and November 2016	<i>epistemic</i> or <i>epistemological</i> and <i>cognition</i> or <i>beliefs</i> and <i>achievement</i> or <i>academic</i> or <i>performance</i> or stems related to these terms	<i>ERIC</i> , <i>PsycINFO</i> , <i>Education Full Text</i> , <i>ProQuest Dissertation & Thesis Abstracts</i> , <i>Web of Science</i> , <i>Citation Index</i>
January and February 2017	<i>epistemic</i> and <i>cognition</i> or <i>beliefs</i> and <i>achievement</i> or <i>academic</i> or <i>performance</i> or stems related to these terms	<i>Google</i> , <i>Google Scholar</i> , <i>ERIC</i> , <i>PsycINFO</i> , <i>Education Full Text</i> , <i>ProQuest Dissertation & Thesis Abstracts</i> , <i>Web of Science</i> , <i>Citation Index</i>
October 2016 and January 2017	Schommer's Epistemological Beliefs Questionnaire, Epistemic Belief Inventory (EBI), Hofer's Domain-specific Epistemological Belief Questionnaire (DEBQ), Indiana Mathematics Belief Scale	<i>Google</i> , <i>Google Scholar</i> , <i>ERIC</i> , <i>PsycINFO</i> , <i>Education Full Text</i> , <i>ProQuest Dissertation & Thesis Abstracts</i> , <i>Web of Science</i> , <i>Citation Index</i>
October 2016	Hofer, Schommer, Schraw, Greene, Bråten, Muis, Chinn	<i>ERIC</i> , <i>PsycINFO</i> , <i>Education Full Text</i> , <i>ProQuest Dissertation & Thesis Abstracts</i> , <i>Web of Science</i> , <i>Citation Index</i>

temic cognition or related constructs and achievement measures but either did not report correlations between the two or did so in a format that could not be converted into our effect size measure. The first authors of these studies were contacted to inquire if they could share the necessary correlations. Most of the authors responded, providing the information needed to determine each study's eligibility for inclusion in the meta-analysis. All of these researchers were also asked to provide any other relevant studies they had conducted but were unpublished.

Given many conference presentations are not listed in the various electronic databases that were used in this investigation, we searched online conference repositories for derivations of the words "epistemic" or "epistemology." The American Educational Research Association (AERA) online paper repository, which goes back to 2010, was scanned and if papers were available, they were downloaded. In two cases, papers were not available through the repository, so authors were contacted but we received no response to these inquiries. All available electronic conference programs over the last 10 years for the American Psychological Association (APA) Division 15, the Association for Psychological Science (APS), the European Association for Research on Learning and Instruction (EARLI), the International Conference of the Learning Sciences (ICLS), the National Association for Research in Science Teacher (NARST), and the International Conferences for "Personal Epistemology and Learning (PEL)" were searched for terms related to the meta-analysis. Seven authors from these conferences were contacted. All responded and three of the studies that were shared with us fit the inclusion criteria of the meta-analysis.

To increase the chances of assembling pertinent unpublished studies, to address publication bias, we employed two strategies. We sent an email through the epistemic cognition listserv (epistemiccognition@listservunc.edu). Additionally, separate emails were sent to 25 researchers who have had multiple publications on epistemic cognition in the past 20 years; these emails were in addition to the separate requests for unpublished studies previously mentioned. Ultimately, three unpublished studies were shared with us and all three were included in the meta-analysis.

Inclusion and Exclusions Criteria

Only studies published in English were considered for inclusion in the meta-analysis. Despite the fact that this criterion created the possibility of language bias, our lack of other-language expertise mandated this choice. Our searches resulted in 737 documents. The second author examined these to determine whether they fit the parameters of the meta-analysis (see Table 3). Forty-eight documents were review, not empirical, articles. The citations in these articles were investigated, but the articles themselves were excluded from the meta-analysis. An additional 105 studies were excluded because we determined they were focused on something other than epistemic cognition (e.g., Angeli & Valanides, 2009; Gupta, 2008). We excluded an additional 278 articles because the authors did not relate epistemic cognition to any academic achievement measure (e.g., Buehl, Alexander, & Murphy, 2002; Peter, Rosman, Mayer, Lechner, & Krampen, 2016). Thirty-six of the studies gathered included epistemic cognition interventions

Table 3
Summary of Document Review for Inclusion in the Meta-Analysis (MA)

Exclusion or inclusion criteria	No. included in the MA	No. not included in the MA
Inclusion: Peer reviewed articles that had correlations between epistemic cognition and academic achievement	100	
Exclusion: Peer reviewed articles that did not examine correlation between epistemic cognition and academic achievement		278
Exclusion: Peer reviewed articles that had relevant measures but not correlations.		47
No responses from authors		11
Data was no longer available		20
Paper was not available in English		1
Authors clarified that results did not fit the meta-analysis criteria		9
No contact information on authors could be located		6
Exclusion: Peer review articles that came up in literature searches, but were review or other pieces, not studies		48
Exclusion: Peer review articles were about interventions		36
Exclusion: Peer review articles in which epistemic cognition was not focus of study		105
Inclusion: Dissertations		
Exclusion: Dissertations	14	35
Dissertations did not have relevant correlations		
Inclusion: Theses	3	
Exclusion: Theses did not have relevant correlations		4
Inclusion: Conference papers	5	
Exclusion: Conference papers		50
Did not have relevant measures and/or correlations		41
Were review or theoretical in nature		9
Inclusion: Conference posters	4	
Exclusion: Conference posters did not have relevant correlations		2
Inclusion: Unpublished	3	
Inclusion: Chapter in book	3	
Total	132	605
Grand total examined		737

(e.g., Kienhues, Bromme, & Stahl, 2008; Sahin, 2010), which went beyond the correlational focus of the meta-analysis. Forty-seven studies that seemed to fit the meta-analysis, but did not include the relevant effect sizes, were excluded for a variety of reasons: we could not find accurate contact information for authors of six of the studies, nine studies did not fit the inclusion criteria of the meta-analysis, 20 studies included data that the authors said was no longer available, 11 researchers that were contacted did not respond to inquiries, and one researcher could only provide relevant information in a language that none of us spoke. Additionally, after in-depth analysis by the second author, 35 accumulated dissertations, four theses, two conference posters, and 41 conference papers were excluded from the meta-analysis for similar reasons (e.g., lacking effect sizes, involving an intervention, author never responded to our inquiry). Further, if dissertations or theses contained data that was published in a subsequent peer-reviewed article, we used only the latter in our study, excluding the non-peer-reviewed source; this ensured we did not include the same data twice in our analyses. All of these investigations led to 132 studies being included in the meta-analysis.

Procedure for Synthesis of Studies

Information retrieved from studies. Five types of information necessary to conduct the moderator analyses were gathered from each study: (a) report characteristics, (b) setting characteristics, (c) sample characteristics, (d) information about the epistemic

cognition or belief construct variable, and (e) information about the academic achievement variable. Table 4 provides a complete list of the information coded for each relevant correlation in a research report.

Effect size estimation. We used the correlation coefficient to estimate the association between the epistemic cognition measure and the student achievement measure. Most studies reported Pearson r values. In two studies, the authors reported Spearman ρ values, and in several studies regression coefficient beta values were listed. For these studies that did not include Pearson r values, we contacted authors to ask if they could share them. If the Spearman ρ values or betas were all that were available, then we used them. An overall effect analysis was run with and without the beta values, and the effect size estimates were very similar; therefore, we kept beta values in the analyses.

One challenge when reviewing epistemic cognition studies is determining the direction of the scales used to measure the phenomenon. In many studies, higher scores on the measure indicated less availing or “naïve” epistemic cognition (e.g., Arslantaş, 2015; Scevak & Cantwell, 2001), whereas for other studies the opposite was true (e.g., Bråten & Strømsø, 2010; Muis et al., 2015). We coded all correlations as reported directly by the studies, but we also coded for Effect Direction, in other words whether higher scores were more or less availing. For analysis, we changed the sign of correlations as necessary to ensure all of them were scaled such that higher scores on the

Table 4
Information Coded in Research Reports

Category	Codes
Report characteristics	1. Author names 2. Year (if published or presented) 3. Report type (i.e., journal article, conference paper, conference poster, book chapter, dissertation, thesis, unpublished study) 4. Abbreviated citation 5. Percent missing data reported
Setting characteristics	1. Location 2. School type (i.e., public, private)
Sample characteristics	1. Group (i.e., elementary school, middle school, high school, college, graduate students, preservice teachers, in-service teachers) 2. Sample size 3. Percent female 4. Percent identifying as white 5. Average age
Epistemic cognition variables	1. Main theoretical model 2. Epistemic cognition construct 3. Epistemic cognition instrument name 4. Reliability of epistemic cognition instrument measure 5. Level of domain-specificity of the epistemic cognition measure (i.e., domain-general, domain-specific, task-specific) 6. Domain/discipline of the study
Academic achievement variables	1. Type of achievement measure (i.e., participant reported, researcher developed, teacher developed, official GPA, standardized test) 2. Academic domain or discipline 3. Type of achievement (i.e., argumentation, general knowledge, conceptual knowledge, declarative knowledge, or procedural knowledge) 4. Reliability of achievement measure
Effect size information	1. Correlation coefficient (i.e., Pearson, Spearman, or beta) 2. p-value 3. Effect direction

epistemic cognition measure indicated more availing or “sophisticated” epistemic cognition.

Coding procedure and reliability. The first two studies included in the meta-analysis were coded for predetermined categories by all three authors independently. Then we met to discuss possible modifications to the coding scheme and confer on any difficulties in coding. Studies 3–16 were coded independently by all three authors again, to measure consistency between coders. At the end of this step, interrater reliability between the first two authors was .86, between the first and third authors it was .77, and between the second and third authors it was .80. Any remaining coding inconsistencies were clarified at this point and the second author created a coding manual to help ensure consistency going forward. The remaining 116 studies were coded by the second author and one of the other two authors, in two steps. First the authors decided how many effect sizes could be coded in each study, then they met to resolve any discrepancies. After this point, all coding was completed independently. The overall interrater reliability between the first and second author was .89 and between the second and third author was .84. The author pairs met after coding and reviewed the coded data and resolved any discrepancies. The first author acted as the arbiter on any unresolved discrepancies. All of the studies and the coded information can be found in the [Supplemental Table](#).

Identification of statistical outliers. Before conducting the meta-analysis, histograms were generated to assess the normality of the distribution of the correlations. After normality was ascertained, the [Grubbs \(1950\)](#) outlier test was conducted. No outliers were identified (using $p < .05$, two-tailed, as the significance level).

Publication bias. Despite our attempts to be as exhaustive in our search as possible, it is possible we did not obtain all studies relevant to our research questions, either due to a failure on our part to effectively search the literature or because some studies, particu-

larly those with statistically nonsignificant results, were never published (i.e., the “file-drawer” problem; [Carson, Schriesheim, & Kinicki, 1990](#)). To explore this possibility, first we created a funnel plot using the Comprehensive Meta-Analysis (CMA; [Borenstein, Hedges, Higgins, & Rothstein, 2014](#)) software (see [Figure 1](#)). The funnel plot was basically symmetrical, indicating an absence of publication bias ([Borenstein et al., 2014](#); [Egger, Smith, Schneider, & Minder, 1997](#); [Mavridis & Salanti, 2014](#)). CMA also allows for the imputing of missing studies. If this changes the appearance of the funnel plot, then publication bias might be the cause. In the case of this meta-analysis, imputing missing studies did not significantly change the appearance of the funnel plot. These findings support that publication bias (i.e., the “file-drawer problem”) was not an issue for this meta-analysis. Additionally, we used the Duval and Tweedie trim-and-fill procedure to test whether the distribution of effect sizes was consistent with variation in effect sizes that would be predicted if the estimates were normally distributed. A skewed distribution could indicate possible publication bias generated by incomplete literature review procedures or by study exclusion criteria used by the authors. No trimming or filling was indicated. Further, we conducted a classic fail-safe N test to determine how many studies with a null effect would be needed to raise the observed p value above .05; that number was 9,265, well above the recommendation of the level at which there is little chance of a significant file-drawer problem ([Carson et al., 1990](#)). We also conducted Orwin’s fail-safe N test, with a trivial correlation set to .100 and a mean correlation in missing studies of .090. This test indicated that an additional 881 studies would be needed to bring the correlation estimate below the trivial level, further arguing against publication bias as a threat to our interpretations of our findings.

Overall effect analyses. In our full analysis of all correlations between epistemic cognition and academic achievement, there were instances where a study had multiple relevant, dependent correlations (e.g., measures of multiple epistemic cognition constructs, each correlated with academic achievement). There are

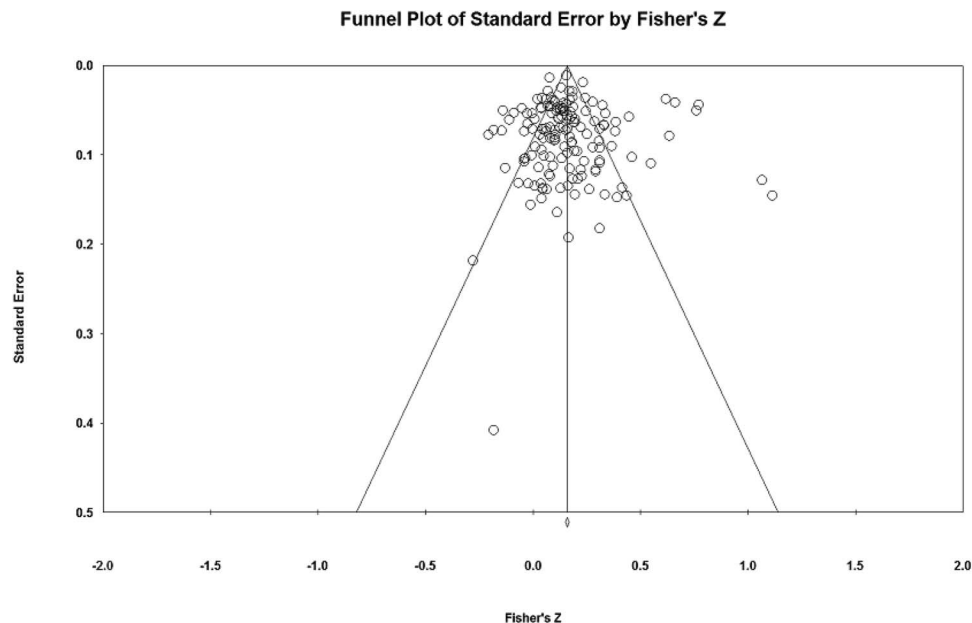


Figure 1. Funnel plot of standard error by Fisher's z .

several ways of dealing with dependent effect sizes in meta-analyses (Tanner-Smith, Tipton, & Polanin, 2016). We chose to run two overall effect size estimate models, one assuming all dependent correlations were perfectly correlated with each other and could be averaged, and another where they were assumed to have a zero correlation (i.e., all effect sizes treated as independent, leading to 752 effect sizes). The two overall effect size estimates, under a random effects model, differed by only .015, and in both models the estimate was statistically significant at the $p < .001$ level. Therefore, we determined there was no need for further adjustments for dependent correlations within a study, and we conducted the overall effect size analysis assuming perfect correlation of within-study correlations, which involved the CMA software calculating the mean of those correlations before analysis. Ultimately, this led to 151 distinct samples, as several of the 132 studies involved multiple groups of students.

Moderator analyses. In the case of the moderator analyses (e.g., epistemic cognition construct), we adjusted for dependent correlations by taking the mean within levels of the moderator. For example, consider a study with two measures of epistemic cognition constructs, such as simple knowledge and certain knowledge scores, and two measures of academic achievement, such as SAT and GPA. For a test of the moderating effect of construct, the correlations between simple knowledge and SAT, and simple knowledge and GPA, would be averaged together, with the same for certain knowledge. For the test of the moderating effect of type of academic achievement, the correlations between simple knowledge and GPA and certain knowledge and GPA would be averaged, with the same procedure being done for correlations between various epistemic constructs and SAT. This “shifting unit of analysis” approach (Cooper, 2017; Dent & Koenka, 2016) was necessary given we had multiple levels of moderators within some studies (e.g., Dai, 2014; Hotulainen & Telivuo, 2014).

Given many of the moderators had some levels that appeared relatively few times, we chose to collapse similar codes together. In some cases, rare codes did not map well onto other ones and were eliminated from moderator analysis. In most moderator categories, levels needed to occur at least six times to be analyzed separately. With the exception of some epistemic cognition models and constructs of particular interest (i.e., epistemic aims, texture and variability), if the sample size was lower than six, then the codes were either combined with other codes or cut from the analysis.

In the Type of Study moderator category, conference posters and conference papers were collapsed into the code “conference proceedings.” Book chapters and peer-reviewed journal articles were collapsed into a “peer-reviewed” code. Dissertations, theses, and unpublished studies were combined into a “not peer-reviewed” code. In the Theoretical Model moderator category, models with similar foci and constructs were combined. Studies grounded in Barzilai and Zohar’s (2014) model were combined with Kuhn’s (Kuhn et al., 2000). Bråten’s (Bråten et al., 2008) model was combined with Greene’s (Greene et al., 2008). Muis et al.’s (2006) TIDE framework was combined with Hofer and Pintrich’s (1997) model. Ryan’s (1984) study, which was originally coded as Perry’s (1970) model but contained a factor similar to simple knowledge, was combined with Schommer’s (1990) model.

In the Group moderator category, one study was eliminated from moderator analysis because the level of school could not be ascertained. Only one study had participants classified as vocational school students, so this category was collapsed into the college student code during moderator analysis. In the Epistemic Cognition Construct moderator category, several constructs were cut from moderator analysis because of low numbers of correlations. Removed constructs included absolutism, multiplism, and evaluativism, as well as empiricism, metaphorism, and rationalism. Other rare codes were collapsed with other codes with similar meanings. Some studies involved a general construct term called “beliefs about knowledge,” which was combined with the simple and certain knowledge code. Another general term, “beliefs about learning” was in some cases combined with constructivism, but in other cases the descriptions in the study led us to collapse this construct with innate ability or simple knowledge. Knowledge is privileged and general nature of learning codes were combined with the omniscient authority code.

In the Epistemic Cognition Instrument moderator category, several measures were cut from analysis because they were used in only a few studies. Think-aloud protocols were only employed in one study so these were eliminated from moderator analysis. Likewise, the Mansell et al. Beliefs about School Learning instrument used in Gonida and Metallidou (2015) and the Learner Model Questionnaire employed in Stoeger (2006) were also cut. The Science Knowledge Questionnaire (Cavallo, Rozman, Blickenstaff, & Walker, 2004), the Psychology-Specific Epistemological Belief Scale, and the instrument employed by Ryan (1984) were deemed similar enough to Schommer’s (1990) EQ, so they were collapsed into that category for moderator analysis. Similarly, the Domain-Specific Epistemological Belief instrument used by Murphy et al. (2010) was included with the Hofer DEBQ, and the Pomeroy Questionnaire used in Tsai (1998) was combined with the Elder (1999) instrument. The last moderator category that had levels that were collapsed was the Domain of Study. Given accounting, design/technology, engineering, Internet, and physical education all had less than six effect sizes related to them, they were combined into an “other” code. All of the studies and these collapsed codes can be found in the Supplemental Table.

Software. GraphPad software was used to conduct the Grubbs’ outlier test. Comprehensive Meta-Analysis software (CMA, Version 3.3.070; Borenstein et al., 2014) was used to calculate all meta-analyses, including moderator analyses.

Results

Our dataset included 132 articles with 752 effect sizes, involving 55,418 participants. Our first research question concerned the overall estimated correlation between epistemic cognition and academic achievement, and was followed by analyses of eight theoretical moderators and three methodological moderators. In all analyses, higher scores on the epistemic cognition measure, as well as the academic achievement measure, indicated more “availing,” “sophisticated,” or “better” performance. Each research question section, other than the section for instrument reliability, includes a table of results, wherein tests of heterogeneity indicate whether the variance of the associated estimate is statistically significantly

different than zero. I-squared values indicate the percentage of that variability that could be attributed to true between-subjects variability.

Overall Correlation Between Epistemic Cognition and Academic Achievement

Fixed effects analyses revealed an overall effect size correlation between epistemic cognition and academic achievement of .158, utilizing 151 studies, 752 unique effect sizes, and 55,418 participants (see Table 5). Random effects modeling produced an overall correlation of .162. Both correlations were statistically significant at the $p < .001$ level, and had overlapping confidence intervals. The variance of these effect sizes was statistically significant, with almost 89% of that variance due to true heterogeneity in the sample. Therefore, these analyses revealed a small (Cohen, 1992), but statistically significant overall correlation between epistemic cognition and academic achievement, which could be better understood via investigations of several theoretical and methodological moderators.

Theoretical Moderators

Model of epistemic cognition. As shown in Table 6, the majority of estimated effect sizes by model were within a fairly narrow range (i.e., .142–.189) except for two with low study counts (i.e., Chinn et al., Muis et al. Royce). The effect sizes for the models in this range were all statistically significant except the one for Kuhn/Barzilai and Zohar, which had only seven sample sizes in its estimate. Estimates with such low study counts should be interpreted with much caution because both the effect size estimate and the confidence interval around that estimate may be untrustworthy. The other notable outlier was constructivism, which had an unusually large effect size estimate.

Epistemic cognition construct. As shown in Table 7, in general, the various construct measures clustered into three tiers. The first tier was comprised of those constructs with statistically nonsignificant effect size estimates, including attainability of truth, structure of knowledge, texture, variability, and epistemic aims. These estimates were based on very few effect sizes each, thus strong interpretations from statistical analyses are not warranted. Omniscient authority had a statistically significant but relatively low correlation with academic achievement, based on 26 samples, thus we put this construct in this first tier as well.

The next tier of constructs, with statistically significant overall effect sizes ranging from .110 to .199, included many of Schommer's (1990) constructs, Hofer's (2000) constructs such as source

of knowledge, as well as Greene et al.'s (2010) justification constructs. Finally, there was a tier of constructs with effect size estimates above .200. These included development of knowledge (Conley et al., 2004), a general justification of knowledge construct (Conley et al., 2004; Hofer, 2000; Strømsø, Bråten, & Samuelstuen, 2008), the multiple justification construct from Bråten et al.'s (2014) work, measures of Kuhn's levels (e.g., Mason & Boscolo, 2004), and those constructs that seemed to combine a wide variety of epistemic cognition ideas, such as when authors summed scores across various epistemic cognition subscales (i.e., General Epistemic Cognition; e.g., Abedalaziz, Yee, Jaladin, Leng, & Baranovich, 2016).

For the majority of epistemic constructs, their effect size 95% confidence intervals (CIs) overlapped with one another, except for those between constructs in the first and third tiers. An interesting exception to this finding was the General Epistemic Cognition construct, which had a wide confidence interval ranging from below .100 to above .400. Some construct estimates were comprised of a low number of effect sizes, but interestingly justification by authority and omniscient authority had low estimates despite substantial sample sizes.

Academic domain. There were few notable differences in overall effect size across various academic domains (see Table 8). All estimates were statistically significant, and all confidence intervals overlapped. The domain of science had the highest estimated effect size, and psychology the lowest. Studies without a strong grounding in domain produced an effect size toward the middle of the distribution within this moderator. The statistically nonsignificant between-levels Q-value for the mixed effects model indicated that there were no detectable differences in average effect size across levels.

Domain-specificity. Studies differed in terms of whether epistemic cognition was measured at the domain-general, domain-specific, or topic-specific level (Muis et al., 2006). The effect size estimate was largest for domain-specific measurements, followed by domain-general, with topic-specific measurements lowest (see Table 9). However, the latter had a relatively small number of effect sizes comprising the estimate, and should be interpreted with caution. All estimates were statistically significant, with overlapping confidence intervals. The statistically nonsignificant mixed effect Q-value indicated that there were no detectable differences in average effect size across levels.

Alignment of domain-specificity. It may be the case that what is more important than the level of specificity of the epistemic cognition measure is the alignment of the specificity of that measure with the specificity of the academic achievement mea-

Table 5
Overall Effect Size Obtained Using Effect Size From All Included Studies

Model	<i>k</i>	<i>ES</i>	95% CI	<i>Z</i>	Test of heterogeneity		
					<i>Q</i>	<i>df</i>	<i>I</i> ²
Fixed effects	151	.158***	[.150, .166]	37.317	1,335.222***	150	88.766
Random effects	151	.162***	[.135, .189]	11.741			

Note. *k* = Number of independent samples; number of effect sizes = 752; total *N* = 55,418. CI = confidence interval.

*** $p < .001$.

Table 6
Theoretical Model Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
Chinn	354	1	.087	[−.017, .190]	1.637	.000		.000
Constructivism	3,922	12	.240**	[.071, .396]	2.757	324.381***	11	96.609
Greene/Bråten	1,490	8	.189***	[.112, .264]	4.740	11.500	7	39.131
Hammer	412	4	.166*	[.006, .317]	2.035	6.778	3	55.737
Hofer & Pintrich	27,664	46	.158***	[.119, .197]	7.766	374.182***	45	87.974
Kuhn	530	7	.142	[−.003, .280]	1.921	13.416*	6	55.277
Muis et al. Royce	160	2	.131	[−.026, .282]	1.639	.238	1	.000
Schommer/Schommer-Aikins	21,055	72	.144***	[.107, .181]	7.484	491.088***	71	85.542
Within levels						1,221.583***	144	
Between levels ^a						3.875/127.423***	7	
Overall	55,587	152 ^b	.152***	[.129, .176]	12.478	1,348.893***	15,1	88.807

Note. CI = confidence interval.

^a Between-levels Q-value entry is the mixed effects value followed by the Fixed effects value. ^b One study had three effect sizes coded as Schommer/Schommer-Aikins model and a separate effect size coded as Hofer & Pintrich model. This led to k = 152.

* p < .05. ** p < .01. *** p < .001.

sure. When classifying studies as either “aligned” (i.e., both the epistemic cognition and academic achievement measure were focused on the same level, either domain-general, domain-specific, or topic-specific) or not, the meta-analysis revealed that aligned studies had higher effect sizes than studies where the measures were not aligned (see Table 10). Both estimates were statistically significant, with overlapping confidence intervals and a statistically nonsignificant mixed effect Q-value, indicating no detectable difference in average effect sizes between levels.

Alignment of study domain. Another question of alignment concerned whether the study context (e.g., a mathematics classroom) and academic achievement measure (e.g., a mathematics knowledge measure) were aligned. Table 11 shows that the overall estimated effect size for aligned studies was higher than the one for nonaligned studies. Both values were statistically significantly different than zero, with confidence intervals overlapping only slightly and a statistically nonsignificant mixed effect between-levels Q value. This suggests that studies with

Table 7
Epistemic Cognition Construct Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
Attainability of truth	2,302	10	.050	[−.001, .100]	1.936	12.421	9	27.539
Certain knowledge	30,983	78	.136***	[.087, .184]	5.423	1,331.492***	77	94.217
Constructivism	13,307	22	.191***	[.098, .281]	3.978	409.932***	21	94.877
Development of knowledge	10,392	18	.274***	[.180, .363]	5.556	322.888***	17	94.375
Epistemic aims	354	1	.087	[−.018, .189]	1.628	.000	0	.000
General epistemic cognition	2,716	14	.266**	[.097, .420]	3.045	251.138***	13	94.824
Innate ability	14,387	52	.137***	[.066, .207]	3.743	934.129***	51	94.540
Justification by authority	2,437	10	.110**	[.041, .177]	3.127	20.019*	9	55.044
Justification of knowledge	12,047	26	.228***	[.160, .294]	6.436	283.136***	25	91.170
Kuhn's levels	294	5	.241***	[.105, .369]	3.429	4.960	4	19.349
Multiple justification	714	7	.225***	[.139, .308]	5.023	7.872	6	23.783
Omniscient authority	13,549	26	.083**	[.024, .141]	2.769	164.407***	25	84.794
Personal justification	2667	13	.169***	[.084, .252]	3.877	49.753***	12	75.881
Quick learning	16,336	52	.199***	[.146, .252]	7.171	590.639***	51	91.365
Simple and certain knowledge	4847	15	.146***	[.089, .203]	4.934	50.710***	14	72.392
Simple knowledge	16,355	52	.162***	[.113, .209]	6.443	466.309***	51	89.063
Source of knowledge	12,467	29	.174***	[.089, .256]	3.969	519.910***	28	94.614
Structure of knowledge	1,282	8	−.035	[−.219, .152]	−.364	69.350***	7	89.932
Texture	111	2	.148	[−.042, .328]	1.529	.670	1	.000
Variability	111	2	.126	[−.065, .307]	1.293	.275	1	.000
Within levels						5,503.630***	429	
Between levels ^a						57.955***/295.810***	25	
Overall	159,319	455	.144***	[.128, .160]	17.525	5,799.440***	454	92.172

Note. Several epistemic cognition constructs have been excluded from this table due to small number of studies including them (i.e., absolutism, evaluativism, metaphorism, multiplism, rationalism). The overall 455 studies include those not listed in the table. CI = confidence interval.

^a Between-levels Q-value entry is the mixed effects value followed by the fixed effects value.

* p < .05. ** p < .01. *** p < .001.

Table 8
Academic Domain Moderator Analysis

Mixed effects analysis	<i>N</i>	<i>k</i>	<i>ES</i>	95% CI	<i>Z</i>	Test of heterogeneity		
						<i>Q</i>	<i>df</i>	<i>I</i> ²
Mathematics	5,935	19	.157*	[.026, .282]	2.350	454.855***	18	96.043
Other ^a	1,706	5	.153*	[.032, .269]	2.478	14.467**	4	72.350
Psychology	1,546	8	.123***	[.062, .182]	3.962	9.440	7	25.851
Public understanding of health and science	2,343	15	.137***	[.081, .191]	4.808	21.602	14	35.191
Science	18,594	49	.199***	[.153, .243]	8.443	389.265***	48	87.669
None/study done outside of domain	26,804	61	.143***	[.107, .178]	7.794	423.353***	60	85.827
Within levels						1,312.982***	151	
Between levels ^b						5.573/39.134***	5	
Overall	56,928	157 ^c	.153***	[.131, .175]	13.301	1,352.115***	156	88.463

Note. CI = confidence interval.

^aThe "Other" category is composed of studies with the domains of accounting, design/technology, engineering, internet, and physical education. ^bBetween-levels *Q*-value entry is the mixed effects value followed by the fixed effects value. ^cThe number of studies (i.e., independent samples) is 157 instead of 150 because several of the studies had multiple domains that were analyzed. To conduct the moderator analysis, these studies had to be re-coded as different subgroups.

* $p < .05$. ** $p < .01$. *** $p < .001$.

aligned epistemic cognition and academic achievement measures produce higher effect sizes than those where the measures are not aligned, but this difference was not statistically detectable.

Type of achievement. Measures of more advanced levels of achievement, such as argumentation or conceptual knowledge, correlated more strongly with epistemic cognition than measures of more basic levels such as declarative or procedural knowledge (see Table 12). General knowledge measures had an estimated effect size in the middle of this distribution. There was significant overlap in confidence intervals among types of measures, however, and there were only two procedural knowledge measures, rendering inferences about that type of achievement moot.

Achievement measure. The strongest overall correlation was between epistemic cognition and standardized test versions of achievement measures (see Table 13). The weakest relation was between epistemic cognition and participant-reported measures. The confidence intervals for each type of achievement measure overlapped, except for these two.

Educational level. Given modern conceptualizations of epistemic cognition as a relevant phenomenon at each educational

level (e.g., Burr & Hofer, 2002; Sandoval et al., 2016), it is not surprising that the overall effect size was statistically significantly different than zero at every educational level (see Table 14). Effect sizes estimates were markedly higher in elementary school, middle school, and mixed populations, which was a somewhat counterintuitive finding given the predominant focus upon college students in the epistemic cognition literature (Hofer & Bendixen, 2012). Overlapping confidence intervals tempered these findings, but the difference between the effect sizes for college and middle school students appears to be robust, given their confidence intervals did not overlap and the finding that the between levels effect was statistically significant.

Methodological Moderators

Peer-review status. As would be expected, peer-reviewed studies had a higher estimated effect size than studies that were not subject to peer review, such as theses, dissertations, or unpublished studies, as well as higher than studies from conference proceedings (see Table 15). Confidence intervals indicated that peer-reviewed studies had a statistically significantly higher effect size estimate than studies in

Table 9
Level of Instrument Specificity Moderator Analysis

Mixed effects analysis	<i>N</i>	<i>k</i>	<i>ES</i>	95% CI	<i>Z</i>	Test of heterogeneity		
						<i>Q</i>	<i>df</i>	<i>I</i> ²
Domain-general	29,722	81	.140***	[.108, .172]	8.556	505.155***	80	84.163
Domain-specific	24,731	67	.184***	[.137, .230]	7.566	827.143***	66	920.021
Topic-specific	1,852	9	.098***	[.046, .149]	3.701	9.602	8	16.684
Within levels						1,341.900***	154	
Between levels ^a						5.881/33.359***	2	
Overall	56,741	157 ^b	.142***	[.119, .166]	11.669	1,375.259***	156	

Note. Several studies had multiple measures, some that were domain-general and others that were domain-specific. CI = confidence interval.

^aBetween-levels *Q*-value entry is the mixed effects value followed by the mixed effects value. ^bThe number of studies (i.e., independent samples) is 158 instead of 150 because several of the studies used multiple epistemic cognition instruments.

*** $p < .001$.

Table 10
Alignment of Instrument-Specificity and Achievement Measure-Specificity Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
Aligned	43,251	110	.176***	[.144, .208]	10.486	1123.473***	109	90.298
Not aligned	15,268	55	.130***	[.089, .171]	6.208	303.796***	54	82.225
Within levels						1427.269***	163	
Between levels ^a						3.030/18.592***	1	
Overall	58,519	165 ^b	.158***	[.133, .183]	12.061	1445.861***	164	88.657

Note. CI = confidence interval.

^a Between-levels *Q*-value entry is the mixed effects value followed by the fixed effects value. ^b The number of studies (i.e., independent samples) is 165 instead of 150 because several of the studies had multiple epistemic cognition measures and achievement measures that were analyzed. To conduct the moderator analysis, these studies had to be re-coded as different subgroups.

*** $p < .001$.

conference proceedings. All effect size estimates were statistically significant.

Reliability. Instrument reliability should positively correlate with overall effect size estimates, because the greater the true score variance, the higher the potential correlations with other measures (Crocker & Algina, 1986). Using metaregression, we found support for this hypothesis [$b = .300$, 95% CI [.181, .420], $p < .001$], along with a statistically nonsignificant intercept term. Given the scaling of instrument reliability from 0 to 1, our metaregression indicated that a theoretical epistemic cognition instrument with a reliability of zero would have a predicted correlation with academic achievement of zero, whereas a theoretical epistemic cognition instrument with a reliability of one would have a correlation of .300.

Instrument. Finally, we expected and found statistically significant differences in the overall effect sizes for the correlation of epistemic cognition and academic achievement by instrument (see Table 16). Focusing upon instruments with a large enough sample of studies to warrant interpretation, we found that Conley et al.'s (2004) questionnaire, the EUI (Kuhn et al., 2000), and the Indiana Mathematics Belief Scale (Kloosterman & Stage, 1992) all had the highest effect size estimates. Notably, Bråten et al.'s (2013) JFK-Q, Schommer's (1990) EQ, and Schraw et al.'s (2002) EBI had lower but still statistically significant estimates, whereas the estimate for Hofer's (2000) DEBQ did not statistically significantly differ from zero.

Discussion

In the last 50 years, there has been a rapid increase in interest in epistemic cognition, with commensurate increase in theoretical and empirical writing on the subject, largely buoyed by arguments that it is key to the kinds of higher level cognitive outcomes and critical thinking necessary for success in the modern world (Greene et al., 2016a). Comprehensive synthetic reviews have done much to capture a picture of research on epistemic cognition, as well as changes in how epistemic cognition has been conceptualized over time and the ongoing debates about those changes (Hofer & Pintrich, 1997; Hofer & Bendixen, 2012; Sandoval et al., 2016). Nonetheless, there remains a need for a systematic meta-analysis of the quantitative empirical evidence regarding how epistemic cognition relates to academic achievement. We conducted such a meta-analysis, and found that the overall relationship between epistemic cognition, primarily involving measures of epistemic beliefs, and academic achievement was small but statistically significant with a relatively large amount of variance, which could be attributed in part to theoretical and methodological moderators. Such findings may not definitively end the debate about the utility of self-report measures of epistemic cognition (cf. Hofer & Sinatra, 2010; Kelly, 2016; Mason, 2016), but they do inform that debate in important ways, and point to a number of empirically supported directions for future research.

Table 11
Alignment of Academic Domain Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
Aligned	38,088	119	.179***	[.145, .211]	10.325	1,199.785***	118	90.165
Not aligned	20,009	44	.111***	[.075, .147]	6.012	183.290***	43	76.540
Within levels						1383.075***	161	
Between levels ^a						7.354**/11.843***	1	
Overall	58,097	163 ^b	.147***	[.122, .171]	11.636	1,394.805***		88.457

Note. CI = confidence interval.

^a Between-levels *Q*-value entry is the mixed effects value followed by the fixed effects value. ^b The number of studies (i.e., independent samples) is 163 instead of 150 because several of the studies had multiple domains that were analyzed. To conduct the moderator analysis, these studies had to be re-coded as different subgroups.

** $p < 0.01$. *** $p < .001$.

Table 12
Type of Achievement Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Tests of heterogeneity		
						Q	df	I ²
Argumentation	1,540	12	.154***	.081 .226	4.108	17.396	11	36.768
Conceptual knowledge	6,249	11	.190***	.073 .301	3.179	74.507***	10	86.579
Declarative knowledge	711	7	.127***	.053 .200	3.365	4.269	6	.000
General knowledge	52,514	126	.160***	.132 .188	10.965	1,197.784***	125	89.564
Procedural knowledge	274	2	.077	-.102 .251	.841	1.927	1	48.100
Within levels						1,295.883***	153	
Between levels ^a						1.766/62.584***	4	
Overall	61,288	158 ^b	.156***	.132 .179	12.549	1,358.468***	157	88.443

Note. CI = confidence interval.

^a Between-levels Q-value entry is the mixed effects value followed by the fixed effects value. ^b The number of studies (i.e., independent samples) is 158 instead of 150 because several of the studies had multiple measures of achievement.

*** $p < .001$.

Across 151 samples, 752 effect sizes, and 55,418 participants, using a mixed effects model, we calculated the overall average correlation between epistemic cognition and achievement to be .162. Interpretation of this correlation effect size can be aided by converting it to a Cohen's (1992) d value, which in this case would be .328, a small to medium effect. This finding suggests that, in theory, an intervention that was successful at moving a person from the lowest levels or forms of epistemic cognition to the highest (e.g., from naïve to sophisticated beliefs in Schommer's [1990] terms), would be associated with change of .328 standard deviations on a measure of academic achievement. For comparison, on Hattie's (2015) list of effect sizes for educational innovations, epistemic cognition would rank just below his preferred $d = .40$ value for practical significance, among other initiatives such as inquiry-based teaching ($d = .31$) and bilingual programs ($d = .37$).

Nonetheless, we caution against using solely this overall effect size estimate to draw a definitive conclusion about the viability of epistemic cognition as a significant predictor of, or lever for, academic achievement. To begin, inferences from empirical research are not solely determined by the magnitude of p values or d values. Any warranted inference is a synthesis of theory, logic, measurement, and statistical evidence. There are strong arguments

that epistemic cognition plays an important role in higher order cognition (cf. Hofer & Bendixen, 2012; Sandoval et al., 2016), and equally strong arguments that there are well-documented measurement issues likely affecting a majority of the studies in this meta-analysis (cf. DeBacker et al., 2008; Mason, 2016). The somewhat small correlation we found in this meta-analysis may be due in no small part to issues of conceptualization and measurement. Indeed, as our analyses showed, there were a number of important moderators of this overall effect size estimate, and investigation of them points to a number of directions for future research that very well might produce stronger measures, and stronger relations, worth pursuing.

Our analyses showed stronger effects for studies coded as falling within particular models of epistemic cognition (e.g., Greene et al., 2008 and Bråten et al., 2011 conceptualizations, Hofer & Pintrich's, model). There was an insufficient number of studies to draw conclusions about some of the other models (e.g., Chinn et al.'s, 2014, AIR model), but it seems warranted to continue pursuing those models given their focus on not only beliefs but also practices, ideals, and processes. Importantly, in our dataset those studies we labeled as being grounded in constructivist theory, as opposed to epistemic cognition, had the highest estimated effect

Table 13
Achievement Measure Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
GPA (official)	4,941	17	.139**	[.035, .240]	2.621	204.313***	16	92.169
Participant reported	4,316	13	.083*	[.018, .147]	2.492	51.884***	12	76.872
Researcher developed	14,621	61	.177***	[.127, .227]	6.820	441.119***	60	86.398
Standardized test	2,642	14	.214***	[.110, .312]	4.005	96.040***	13	86.464
Teacher developed	36,919	59	.159***	[.123, .194]	8.623	609.541***	58	90.485
Within levels						1,402.897***	159	
Between levels ^a						7.028/50.535***	4	
Overall	63,439	164 ^b	.154***	[.129, .179]	11.956	1,453.432***	163	88.785

Note. CI = confidence interval.

^a Between-levels Q-value entry is the mixed effects value followed by the fixed effects value. ^b The number of studies (i.e., independent samples) is 164 instead of 150 because several of the studies had multiple measures of achievement.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 14
Level of Schooling Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
Elementary	1,647	8	.212***	.114 .305	4.192	26.963***	7	74.038
Middle school	2,382	10	.246***	.172 .317	6.329	25.019**	9	64.027
High school	27,443	39	.175***	.127 .222	7.050	493.598***	38	92.301
College	15,312	68	.131***	.097 .164	7.533	273.894***	67	75.538
Graduate	289	4	.164*	.031 .291	2.416	3.724	3	19.449
Mixed	6,256	11	.233**	.084 .372	3.039	335.350***	10	97.018
Preservice teachers	2,065	10	.111	-.022 .240	1.641	68.948***	9	86.947
Within levels						1,227.497***	143	
Between levels ^a						15.111*/107.726***	7	
Overall	55,418	151	.159***	.136 .183	12.976	1,335.222***	150	88.766

Note. The one study that had unidentified student group is not included in the individual analysis. CI = confidence interval.

^a Between-levels Q-value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

size. This suggests that such studies may be skewing impressions of the literature. Of course, there is a confound between models of epistemic cognition and the constructs posited within them, therefore it is not surprising that constructs we labeled as constructivist in nature also had a relatively high effect size estimate. There are arguments against treating constructivism as a kind of epistemic cognition, because it concerns how humans create mental representations, rather than how they determine what is and is not knowledge qua knowledge (Greene et al., 2016a; Sandoval et al., 2016). This suggests that researchers should be cautious when deciding how to define and measure epistemic cognition, and be wary of making inferences about the utility of epistemic cognition from studies involving constructivist operationalizations.

On the other hand, the constructs associated with Kuhn et al.'s (2000) work, as well as constructs related to the development and justification of knowledge, proved some of the stronger predictors of academic achievement in our analysis. Perhaps most interesting was that constructs associated with participants' views of authority (e.g., omniscient authority, justification by authority) had some of the lowest estimated effect sizes. Philosophical epistemologists have placed a great deal of emphasis on how people interpret and use knowledge claims from testimony (i.e., authority), and the burgeoning field of multiple source use (Braasch, Bråten, & McCrudden, in press) includes a growing research emphasis on how

people evaluate and use various kinds of sources (Rouet, Britt, & Durik, 2017). It may be that critiques regarding the adequacy of Likert-type measures of epistemic cognition should be most heavily levied against items that treat appeals to authority in an overly simplistic way, either as always availing, or always not availing (cf. Chinn et al., 2011; Greene & Yu, 2014; Hofer & Sinatra, 2010). There is no one authority, and items or measures that treat authority as a monolithic, good or bad source of knowledge are unlikely to result in accurate estimates of how people actually evaluate and use sources of testimony.

The nature of learning factors (e.g., quick learning and innate ability; Schommer-Aikins, 2004) did not have demonstrably larger relations with academic achievement than other constructs typically seen as more "epistemic" in nature (Greene et al., 2008; Hofer & Pintrich, 1997; Sandoval, 2009). This finding suggests that the inclusion of those factors in studies of epistemic cognition and academic achievement may not be biasing interpretations of the strength of that relationship, but clarity and precision concerns may still warrant treating those constructs as different in kind than other aspects of epistemic cognition (Alexander, 2016).

We were surprised to find the highest estimated effect size for "general" measures of epistemic cognition, which was comprised primarily of those measures involving summed scores across theoretically distinct subscales or constructs. However, the rather

Table 15
Peer Review Status Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
Conference proceeding	2,400	10	.097***	[.056, .138]	4.599	9.263	9	2.834
Not peer reviewed	7,146	27	.132**	[.049, .213]	3.104	302.155***	26	91.395
Peer reviewed	45,872	114	.175***	[.145, .204]	11.397	1,009.941***	113	88.811
Within levels						1,321.359***	148	
Between levels ^a						9.345**/13.863***	2	
Overall	55,418	151	.146***	[.123, .169]	12.302	1,335.222***	150	88.766

Note. Conference proceedings comprise conference posters and conference papers. The "not peer reviewed" category is composed of theses, dissertations, and unpublished studies.

^a Between-levels Q-value entry is the mixed effects value followed by the fixed effects value.

** $p < .01$. *** $p < .001$.

Table 16
Epistemic Cognition Instrument Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI	Z	Test of heterogeneity		
						Q	df	I ²
Bråten JFK-Q	1,439	7	.179***	[.100, .255]	4.411	9.881	6	39.276
CEASBQ	354	1	.087	[−.017, .190]	1.637	.000	0	0.000
Conley et al. Q	10,651	17	.242***	[.155, .325]	5.327	27.192***	16	94.012
CAEB	179	3	.170*	[.022, .312]	2.242	.764	2	0.000
EBAPS	80	1	.028	[−.193, .246]	.243	.000	0	0.000
Elder Q	412	4	.166*	[.006, .317]	2.035	6.778	3	55.737
Epistemic thinking assessment	231	2	.006	[−.124, .136]	.094	.424	1	0.000
Epistemic understanding inventory	299	5	.242***	[.104, .371]	3.399	5.038	4	20.608
GEBEP	76	1	.284*	[.062, .478]	2.491	.000	0	0.000
Hofer DEBQ	5,942	14	.059	[−.016, .133]	1.531	75.769***	13	82.842
Hofer's nature of mathematics epistemological belief scale	438	1	.040	[−.054, .133]	.836	.000	0	0.000
Indiana mathematics belief scale	2,495	5	.421***	[.196, .604]	3.509	160.402***	4	97.506
Jehng et al. Questionnaire	2,122	5	.051	[−.005, .107]	1.771	5.932	4	43.572
Mathematics epistemological beliefs scale	967	5	.035	[−.158, .226]	.352	37.401***	4	89.305
Muis/Royce psycho-epistemological profile	160	2	.131	[−.026, .282]	1.639	.238	1	0.000
Schommer EQ	17,438	53	.156***	[.115, .195]	7.503	344.237***	52	84.894
Schraw et al. EBI	3,095	17	.154**	[.044, .259]	2.748	145.326***	16	88.990
Scientific epistemological views scale	145	1	.099	[−.065, .258]	1.180	.000	0	0.000
Topic-specific epistemic beliefs	1,265	6	.125***	[.051, .198]	3.295	7.807	5	35.955
Wood & Kardash EBS	594	3	.148*	[.067, .226]	3.579	.082	2	.000
Within levels						1,081.809***	137	
Between levels ^a						47.775***/291.030***	20	
Overall	564,912	158 ^b	.132***	[.112, .152]	12.783	1,372.839***	157	88.564

Note. JFK-Q = Justification for Knowing Questionnaire; CEASBQ = Context-Specific Epistemic Aims and Source Beliefs Questionnaire; CAEB = Connotative Aspects of Epistemological Beliefs Questionnaire; EPABS = Epistemological Beliefs Assessment for Physical Science; GEBEP = Greek Epistemological Beliefs Evaluation Instrument for Physics; CI = confidence interval.

^a Between-levels *Q*-value entry is the mixed effects value followed by the fixed effects value. ^b The number of studies (i.e., independent samples) is 158 instead of 150 because several of the studies used multiple epistemic cognition instruments.

* $p < .05$. ** $p < .01$. *** $p < .001$.

large amount of variance around this effect size (i.e., 95% CI [.097, .420]) is telling; it suggests that these measures vary greatly in their predictive utility. Much more research is needed to fully understand this finding, but there is a strong clue toward an explanation for this effect in our finding that instrument reliability positively and statistically significantly correlates with estimated effect size. Instruments with more items tend to have higher reliability (Crocker & Algina, 1986) and “general” measures of epistemic cognition tended to have more items due to summing across subscales. Therefore, it is reasonable to presume that the relatively large effect size estimated for “general” measures of epistemic cognition may simply be a function of their higher reliability.

Arguments for the validity of inferences from scores on a measure depend upon numerous factors, with instrument reliability being a necessary but not sufficient aspect of that argument (Crocker & Algina, 1986). Therefore, researchers interested in finding the best measure of epistemic cognition for their study, as well as those interested in developing their own measure, would be wise to consider our other moderator findings as well. One of the most prevalent conceptual debates in the field of epistemic cognition has centered around issues of domain-specificity. Our meta-analysis results showed statistically significant relations between epistemic cognition and academic achievement at domain-general, domain-specific, and topic-specific levels, with the highest for domain-specific instruments. This argues in favor of conceptualizations of epistemic cognition that span each of those levels (e.g.,

Hofer & Pintrich, 1997; Muis et al., 2006), and further suggests that researchers should consider including multiple measures of epistemic cognition, across these levels, in their studies. Alignment of measures of epistemic cognition and academic achievement seems to be another important factor when studying the relation between these constructs. Studies aligned in terms of domain-specificity, as well as those aligned in terms of academic domain, had higher correlations than nonaligned studies.

Furthermore, effect size estimates were higher when the academic achievement measure captured conceptual knowledge or argumentation, as opposed to declarative or procedural knowledge. Conceptual knowledge and argumentation achievement measures capture the kinds of higher order processing posited to require epistemic cognition (Greene & Yu, 2016; Sinatra & Chinn, 2011), and the higher effect size estimates support such claims. These findings may be expected, but they are important. They bolster arguments that researchers should take domain differences in argumentation, conceptual understanding, and epistemic cognition seriously, and consider tailoring their instruments to the kinds of epistemic practices, epistemic ideals, and reliable processes considered normative in particular domains (Chinn et al., 2011; Sandoval, 2012; Sandoval et al., 2016; Weber et al., 2014). True conceptual understanding, and the ability to engage in argument about it, likely requires deep knowledge of how knowledge develops within domains, including the epistemic practices used to create and evaluate those claims (Duschl, 2008; NGSS Lead States, 2013). Likewise, researchers looking to justify their inves-

tigation of epistemic cognition, and educators wishing to argue for emphasizing epistemic cognition in their curricula, should focus on its relations with the kinds of achievement likely to be sensitive to such thinking, as opposed to basic declarative knowledge.

Finally, we found that epistemic cognition was a statistically significant predictor of academic achievement from elementary school through graduate school. Again, these findings support more modern arguments that even young children can and do critique and use knowledge, and that variance in such abilities plays a role in their educational success (Bendixen & Feucht, 2010; Bråten, 2016; Sandoval et al., 2016). The overlapping confidence intervals among the various levels of schooling preclude inferring that epistemic cognition is more strongly related to academic achievement in one level than another. However, the noticeably larger effect size estimates in elementary and middle school samples, compared with college, strongly argue for additional research in these populations. Our data were not amenable to testing the proposed order or progression posited in developmental models (e.g., Kuhn et al., 2000), nor could they answer questions about whether such development begins first in social sciences as compared with hard sciences (Chandler et al., 2002). These issues are two of many that will require further research, with new measures hopefully informed by the lessons of this meta-analysis.

Limitations

This meta-analysis is necessarily constrained by several factors. To begin, despite our earnest attempts to include all relevant empirical research, it may be the case that studies were missed, or not accessible to us. Our analyses would suggest no tangible effect of the “file-drawer” problem, but it is possible there was one. Further, this was a meta-analysis of nonexperimental studies, and therefore no causal relations can be inferred. Measurement problems continue to plague the field of epistemic cognition, likely attenuating the relationships between epistemic cognition and achievement, thus there is the threat of a “garbage in, garbage out” problem in these data. However, with such a large number of studies and participants, and such low *p* values for our correlation effect size estimates, it seems unlikely that there is in actuality no relationship between epistemic cognition and achievement; on the other hand, the relatively low reliabilities for most epistemic cognition measures may suggest its true relationship with academic achievement may be underestimated in these data.

Another limitation of our study was, despite the large sample size, due to sparse data we were unable to conduct a metaregression with all of our moderators to determine which were most predictive of variance in effect size estimates. There was a great deal of confounding across moderators such as model and construct, as well as disparate sample sizes across models (e.g., Schommer’s model formed the basis of 72 studies, whereas Hammer’s model was the foundation for only 4), constructs, and other moderator levels. As researchers conduct further research on some of the newer models (e.g., Chinn et al., 2011), it may be possible to conduct such metaregressions.

Finally, this meta-analysis, with a necessary focus on quantitative empirical research, is necessarily bounded by critiques of the underlying assumptions of such work. Researchers arguing for situated views of epistemic cognition (Chinn et al., 2011; Elby & Hammer, 2010; Sandoval, 2012), and those who question the

viability and accuracy of self-report measures of epistemic cognition (Hofer & Sinatra, 2010; Kelly, 2016; Mason, 2016), share similar, and legitimate, concerns about whether the data used in our meta-analysis reveal much at all about how people actually enact epistemic cognition in authentic settings and practice (Sandoval, 2005). Studies showing expected changes in self-reported epistemic cognition over the course of experiences posited to have such effects (e.g., Muis & Duffy, 2013) may be revealing actual change, or they may simply be capturing participants’ growing understanding of formal epistemology, rather than the adoption of more adaptive practical epistemology (Sandoval, 2005). Evidence supporting the latter would also serve to limit the utility of the results of this meta-analysis.

Future Directions

Limitations both actual and hypothetical notwithstanding, our findings reveal a number of promising directions for future research. Clearly, one important next step would be to do a meta-analysis of the experimental studies involving epistemic cognition and academic achievement (e.g., Muis & Duffy, 2013). The findings from our meta-analysis suggest that instrument reliability is a key determiner of relations between epistemic cognition and other constructs, therefore studies of interventions designed to change epistemic cognition must include instruments that can capture such effects. Likewise, our findings suggest such interventions are more likely to affect performance on conceptual and argumentation measures, as opposed to declarative knowledge ones.

Our moderator analyses suggest a great deal of variance in the predictive utility of various epistemic cognition instruments. Many researchers have argued for more thorough and long-term testing and refinement of such instruments (e.g., Greene & Yu, 2014), and our findings support those arguments. Instrument development is often a long and arduous process (Crocker & Algina, 1986), and it may be that epistemic cognition researchers would be better served focusing their near-term efforts on refining their measures, as opposed to using current versions to predict other phenomena of interest. However, we must acknowledge that such efforts may have limited efficacy. There have been numerous compelling arguments against self-report measures of epistemic cognition (e.g., Hofer & Sinatra, 2010; Kelly, 2016; Sandoval, 2012), and it may be the case that even the most reliable instrument, focused on the most predictive constructs, in the most amenable of settings, may not truly capture all of the relevant phenomena. As Sinatra (2016) has argued, epistemic cognition most likely includes not only attitudes or beliefs such as whether knowledge is simple or certain, but also particular cognitive, metacognitive, and epistemic processes (Barzilai & Zohar, 2014; Chinn et al., 2011; Greene et al., 2016b). As Sandoval (2005, 2012) has argued, task-based observations may be needed to truly understand how people enact their epistemic cognition. More research is needed, but it may be the case that understanding what people believe or what they can articulate about knowledge and knowing (e.g., epistemic aims, epistemic beliefs), is but one piece of a larger picture of epistemic cognition that includes the kinds of epistemic practices they enact, the kinds of reflection they engage in during those practices, and the ways their context affords or constrains such thoughts and actions (Sandoval et al., 2016). Researchers should consider administering measures of epistemic beliefs before, during, and after

actual tasks involving knowledge and knowing, and collecting other online forms of data to assess epistemic cognition in action (e.g., think-aloud protocols; Greene et al., 2014).

Such views argue for increased conceptualization and research on the situated nature of epistemic cognition. This is a reasonable and promising direction, as well. A complex phenomenon such as epistemic cognition almost certainly emerges, in part, via complex interactions among people, contexts, and the sociohistorical tools and practices relevant to participation in those contexts (Turner & Nolen, 2015). Understanding and modeling epistemic cognition certainly requires such a perspective, with the hope that findings from a situative perspective could inform recommendations on how educators can create contexts amenable to the development and internalization of adaptive epistemic cognition, and the various higher order processes contingent upon it, such as critical thinking (King & Kitchener, 1994; Greene & Yu, 2016). The results from this meta-analysis can be used to inform such educational efforts. Epistemic cognition appears to be (a) relevant from elementary school through graduate school, (b) correlated with both local measures of GPA and standardized measures of conceptual understanding, argumentation, and achievement, and (c) varied in its effect across disciplines, domains, and topics. Such findings argue strongly for continued research in this area.

Conclusion

In sum, our meta-analysis of the nonexperimental empirical literature on the relationship between epistemic cognition and academic achievement revealed a small but potentially powerful relationship in need of further research (Alexander, Murphy, Guan, & Murphy, 1998; Greene et al., 2016b). Our moderator analyses revealed a number of statistically and practically significant factors explaining a wide disparity in the strength of this relationship. Researchers interested in quantitative research on epistemic cognition should consider the results of this work when choosing and developing their instruments, so that the most accurate estimate of epistemic cognition's role in higher order thinking can be determined. It seems unwise to continue uncritically using self-report measures of epistemic cognition; more psychometric development work is needed. Calls for changes in educational and curricular standards to prepare students with 21st century skills (Council of Chief State School Officers, 2010; NGSS Lead States, 2013) must be met with research on how best to achieve those goals, and our findings continue to support epistemic cognition as one key aspect of such work.

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Correction to Greene et al. (2018)

In the article “A Meta-Analytic Review of the Relationship Between Epistemic Cognition and Academic Achievement,” by Jeffrey A. Greene, Brian M. Cartiff, and Rebekah F. Duke (*Journal of Educational Psychology*, Advance online publication. March 8, 2018. <http://dx.doi.org/10.1037/edu0000263>), Table 7 contained a production-related error. Overall *N* was listed as “1,9,319” when it should be “159,319.” All versions of this article have been corrected.

<http://dx.doi.org/10.1037/edu0000312>